ANALYSTLAB AFRICA

Experience Lab Internship Programme

Official Weekly Assignment

Week 1: ShopSmart Retail Analytics & AI Implementation Project

Document Control

Programme	AnalystLab Africa Experience Lab
Week	Week 1
Project	ShopSmart Retail Analytics & AI Implementation
Project Title	Delivering a Retail Analytics and Artificial Intelligence Solution for ShopSmart Retail Group
Version	1.0
Issue Date	_____
Submission Deadline	End of Week 1 (Sunday, 11:59 PM WAT)
Submission Method	Official AnalystLab Africa Week 1 Submission Form

Part 1: Data Understanding: Report & Schema Architecture

1. Number of Datasets Provided

A total of **10 datasets** (CSV files) were provided for this analysis, consisting of 9 core operational tables (Customers, Orders, Order Items, Products, Stores, Employees, Inventory, Returns, and Suppliers) along with a supporting Data Dictionary used for field definitions and reference.

2. Dataset Representations

- **Customers (ALA_ShopSmart_Customers)**: Stores individual customer profiles, including demographic details (Name, Gender, Age) and contact information.
- **Orders (ShopSmart_Orders)**: Records transactional headers detailing purchases made, including order dates, quantities, and associated customer and store references.
- **Order Items (ShopSmart_Order_Items)**: Captures line-item details for each order, including specific products purchased, pricing, quantities, and discounts.
- **Products (ShopSmart_Products)**: Houses master catalog data covering product categories, sub-categories, brands, pricing, and supplier source mapping.
- **Stores (ShopSmart_Stores)**: Contains physical retail store locations, operational regions, states, cities, and opening dates.
- **Employees (ShopSmart_Employees)**: Manages staff profiles, roles, hire dates, salaries, and store alignments.
- **Inventory (ShopSmart_Inventory)**: Tracks stock levels, reorder thresholds, and restock schedules per product and store location.
- **Returns (ShopSmart_Returns)**: Logs post-purchase product returns, including return dates, reasons, and refund amounts.
- **Suppliers (ShopSmart_Suppliers)**: Details vendor contact information, locations, and supply chain origins.
- **Data Dictionary**: Acts as a metadata reference manual outlining field types and definitions (excluded from database import).

3. Primary Key (PK) Identification

Each table relies on a unique identifier column as its primary key to ensure row uniqueness:

- **Customers**: Customer_ID
- **Orders**: Order_ID
- **Order Items**: Order_Item_ID
- **Products**: Product_ID
- **Stores**: Store_ID
- **Employees**: Employee_ID
- **Inventory**: Inventory_ID
- **Returns**: Return_ID
- **Suppliers**: Supplier_ID

4. Foreign Key (FK) Identification

Foreign keys connect child tables back to parent master tables to maintain relational integrity:

- **Orders**: Customer_ID (links to Customers), Store_ID (links to Stores)
- **Order Items**: Order_ID (links to Orders), Product_ID (links to Products)
- **Employees**: Store_ID (links to Stores)
- **Inventory**: Store_ID (links to Stores), Product_ID (links to Products)
- **Returns**: Order_ID (links to Orders)
- **Products**: Supplier_ID (links to Suppliers)

5. Table Relationships & ER Diagram Summary

The tables are integrated into a relational database management system (Microsoft Access) featuring enforced referential integrity across all 1-to-Many (1:∞) relationships:

- **Customers to Orders** (1:∞): One customer places multiple orders.
- **Stores to Orders / Employees / Inventory** (1:∞): One store manages multiple orders, staffs multiple employees, and stocks multiple inventory items.
- **Products to Order Items / Inventory** (1:∞): One product appears across multiple order line items and store inventories.
- **Suppliers to Products** (1:∞): One supplier provides multiple products.
- **Orders to Order Items / Returns** (1:∞): One order contains multiple line items and can trigger return events.

Below is a diagram designed by Microsoft access illustrating the database relationships

